

Gap-Fuzzy Adaptive Signal Control: Enhancing Urban Traffic Efficiency

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Abstract—Traffic congestion in urban areas has intensified due to the rapid growth of vehicles, inadequate infrastructure planning, and unsynchronized traffic signals. This study presents an adaptive traffic signal control strategy called Gap-Fuzzy, which combines the Mamdani fuzzy logic controller with a gap-out detection mechanism. The system dynamically adjusts the duration of the green light based on real-time traffic data, including queue length and arrival rate. Furthermore, the green phase is terminated early if prolonged gaps in vehicle flow are detected. We evaluated the proposed Gap-Fuzzy system using the SUMO microscopic traffic simulator. The results indicate that it reduced vehicle waiting times by up to 70% and CO₂ emissions by 25% compared to fixed-time control. In addition, it outperformed the SUMO actuated controller under low, medium, and high traffic conditions while maintaining comparable performance under very high demand. These findings highlight the effectiveness of integrating fuzzy logic with gap-out detection to enhance traffic flow and minimize environmental impact.

Index Terms—Intelligent traffic control systems (ITCS), fuzzy logic, gap-out inference, SUMO

I. INTRODUCTION

Urban centers are facing increasing levels of traffic congestion, particularly in historic and densely populated areas. This problem is driven by several factors, including the rapid growth in vehicle ownership, inadequate urban planning, poor coordination of traffic signals, and lack of a traffic culture.

In this context, traffic signal control plays an essential role in the effective management of urban transportation systems, with the objective of optimizing vehicle flow.

Traditional fixed-time traffic signals often fail to adapt to fluctuating traffic conditions, which can lead to long waiting times that negatively affect user experience. However, dynamic control strategies, such as actuated or adaptive signal systems, adjust traffic light phases based on real-time traffic data. These approaches aim to improve traffic efficiency by minimizing

delays, reducing vehicle queues, and responding to varying demand patterns. However, their performance heavily depends on the quality and reliability of sensor data, as well as the ability to make timely and accurate decisions under uncertainty.

Recent approaches incorporate fuzzy logic techniques for context-aware signal control, allowing real-time adjustments of signal cycles based on current traffic conditions. In addition, the gap-out detection mechanism enables the control system to end a green phase early, before reaching the maximum green time, if no vehicles are detected in the active lane.

In this work, we propose a traffic signal control system, termed Gap-Fuzzy, which integrates a Mamdani-type fuzzy logic controller with a gap-out inference mechanism. The system continuously monitors vehicle queues and arrival rates to dynamically adjust green phase durations in real time. The gap-out logic enables the early termination of the green phase when no vehicles are detected in the active lane, thereby preventing under-utilization of intersection capacity. This approach aims to improve traffic mobility in critical urban areas, reduce waiting times during peak periods, and mitigate the environmental impact of congestion, particularly in terms of vehicle emissions.

The proposed Gap-Fuzzy traffic control system leverages vehicular communication technologies, together with data collected from cameras, loop detectors, and fiber optic sensors, to improve the accuracy and responsiveness of traffic monitoring and control; see Fig. 1. The system relies on vehicle-to-everything (V2X) communication, which includes vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I), as well as sensor data transmitted to roadside units (RSUs) that are integrated with traffic signal controllers. This architecture enables real-time data exchange among connected vehicles, sensors, cameras, and the intersection controller. The continu-

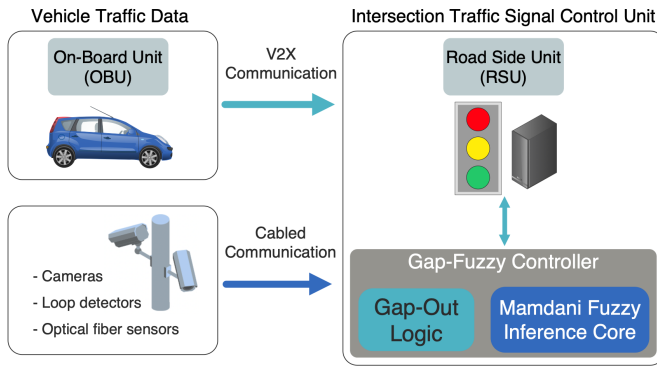


Fig. 1: Gap-Fuzzy traffic signal control architecture

ous flow of traffic data facilitates reliable vehicle counting and accurate estimation of queue lengths and arrival rates, which are essential inputs for both the Mamdani fuzzy inference system and the gap-out inference core, as illustrated in Fig. 1.

To evaluate the effectiveness of the proposed Gap-Fuzzy traffic control system, we perform a comparative analysis against the traffic light control modes available in SUMO [1]. SUMO is a widely used microscopic traffic simulator that supports various control strategies, including

- *Static control mode*: Operates with fixed phase durations of the signal, regardless of traffic demand. This mode reflects the most common real-world configuration implemented in urban intersections.
- *Actuated control mode*: Using simulated inductive loop detectors placed at predefined distances from the intersection. This mode adaptively adjusts the duration of the green phase within the specified minimum and maximum limits based on real-time traffic conditions. The controller monitors the time gaps between successive vehicles: if vehicles are detected in close succession, the green phase is extended; if the gap exceeds a defined threshold (max-gap), the controller terminates the current phase and switches to the next. This approach represents a data-driven and adaptive traffic control strategy.

The main contributions of this work are summarized as follows:

- Integration of fuzzy logic with gap-out detection mechanism to improve real-time response to traffic dynamics. This hybrid approach allows real-time signal timing adjustments based on queue length and arrival rate, while preventing green phase prolongation when no vehicles are present.
- Reduction of vehicle waiting times and underuse at intersections. The proposed system significantly reduces vehicle delays by terminating green phases early during periods of low demand.
- An in-depth evaluation of the proposed system in comparison with the default static timing and actuated control modes available in the SUMO traffic simulator.

The remainder of this paper is structured as follows. Section II reviews and compares related work. Section III describes

the components of the system and the overall functionality. Section IV evaluates the performance of the system and analyzes the environmental impact of integrating the Gap-Fuzzy mechanism. Finally, Section V presents the conclusions and outlines the direction for future work.

II. RELATED WORK

To overcome the limitations of traditional fixed-time traffic signal control, intelligent traffic control systems (ITCS) are increasingly utilizing advanced techniques such as machine learning, fuzzy logic, reinforcement learning, and hybrid approaches.

Table I presents a comparative overview of recent studies in the field of ITCS, highlighting the diversity of control strategies, input parameters, and evaluation metrics used. Reviews of studies can be broadly categorized into three groups: machine learning-based systems, fuzzy logic-based systems, and hybrid systems that integrate elements from multiple methodologies.

A. Machine learning-based control

In [2], the authors introduce a heuristic algorithm aimed at adjusting multimodal traffic patterns in the city of Berlin. The main goal of the algorithm is to optimize the traffic signal cycles based on real-time traffic frequency. The methodology involves conducting simulation experiments that incorporate manual adjustments to the road network, such as adding lanes, rerouting traffic, and relocating traffic signals with fixed time intervals. The results show slight improvements under low traffic conditions, more significant gains under medium traffic, and limited effectiveness during periods of high traffic, as measured by metrics such as waiting time and travel time.

The authors of [3] introduce an intelligent traffic light control system that takes into account both vehicles and pedestrians. This system classifies traffic volumes into four categories to later use machine learning models to predict optimal traffic light times. A CNN is used to categorize traffic, while Artificial Neural Networks (ANN) and Support Vector Machines (SVM) are used to determine appropriate durations for green lights. The primary goal is to minimize delays for all road users. The proposed logic is implemented and tested on the SUMO simulator, showing a reduction in waiting time from 12% to 27% for vehicles and from 9% to 23% for pedestrians compared to traditional fixed-time and semidynamic traffic control methods.

In [4], the authors present an intersection control algorithm aimed at reducing emissions at stop lines while also evaluating the electricity consumption of electric vehicles. Their approach is based on a reinforcement learning (RL) framework, where vehicles are treated as agents, and the traffic infrastructure acts as the environment in which they operate. The proposed architecture focuses on balancing efficiency and privacy. Vehicle sensor data are processed locally to protect sensitive information, while the traffic infrastructure only shares public information, such as traffic light cycles. This design highlights the potential of RL-based strategies not

only to improve environmental outcomes but also to address data privacy concerns in intelligent transportation systems.

B. Standalone fuzzy logic control

The fuzzy logic technique is widely used in control systems to manage complex, dynamic, and uncertain environments. Its ability to model expert knowledge through linguistic rules enables the design of more flexible and adaptive control systems. This approach is also used to control traffic signals, which shows improvements in traffic flow efficiency.

In [5], the authors present a fuzzy logic-based adaptive traffic signal control strategy that aims to prevent queue spillover at isolated intersections. The approach introduces an overflow index to anticipate downstream congestion risks resulting from bottlenecks. This index, combined with real-time measurements of queue length and red phase duration, feeds into a fuzzy inference system that dynamically adjusts the allocation of green time. The system is evaluated using the SUMO simulator, with performance measured by maximum queue length and average vehicle delay.

In contrast, [6] proposes a multi-intersection traffic control framework that combines fuzzy logic with a rule base reduction technique to mitigate computational complexity. This agent-based system incorporates multiple inputs, including vehicle queue length, waiting time, distance to the intersection, and presence of emergency vehicles, allowing adaptive phase selection. The evaluation, also performed in SUMO, considers metrics such as average waiting time, fuel consumption, and CO₂ emissions. Both studies demonstrate the flexibility of fuzzy logic for adaptive traffic control while targeting different levels of network granularity and operational objectives.

In [7], the authors discuss the design and simulation of both fuzzy and conventional controllers for a two-intersection network experiencing overloaded traffic conditions. The fuzzy controller employs heuristic rules and was compared with a proportional analytical approach through simulations carried out in MATLAB and Aimsun. Although both controllers exhibited similar performance at individual intersections, the fuzzy controller delivered superior overall results.

The work presented in [8] introduces an Adaptive Traffic Signal Control (ATSC) system based on Model Predictive Control (MPC), which utilizes real-time data from connected and automated vehicles (CAV) as probes. This approach incorporates dynamic road segmentation to estimate traffic conditions and employs a Store-and-Forward Model (SFM) to predict queue lengths. To reduce computational complexity, a Linear Quadratic Regulator (LQR) is implemented. The overall system performance is evaluated by comparing with that of fixed-time and traditional MPC methods in the SUMO simulator. The proposed system achieved a total delay reduction of up to 38.5% and an average queue length reduction of 34.5%.

Fuzzy logic is a technique that has a long-standing track record in the field of traffic control. There are recent notable examples using simulations based on real-world street layouts. However, most of these studies focus solely on a single intersection with high traffic volume. In this work, we initially

propose the implementation of an independent control system for each traffic light, based on fuzzy logic, and adapted to traffic flow in a dense urban scenario. This allows for a direct comparison with the classical system currently used in real-life traffic management.

C. Hybrid Traffic Control System (HTCS)

In [9], the authors present a HTCS called FuzzyLight, a two-stage framework based on fuzzy logic. In the first stage, fuzzy logic is used to select the optimal signal phase by taking into account both the lengths of the queue and the counts of vehicles. In the second stage, the system determines the duration of the signal phase by combining fuzzy logic with RL. The two-stage design focuses on robustness against sensor noise and instability in RL environments. To evaluate the effectiveness of the proposed controller, throughput and the number of stops are considered.

In studies [10] and [11], the authors propose HTCS that integrate fuzzy logic with machine learning techniques. In the first study [10], a Random Forest classifier is combined with fuzzy inference to predict traffic states and dynamically adjust signal phases based on real-time conditions. In the second study [11], the authors incorporate deep reinforcement learning (DRL) with fuzzy logic, where the reward function is designed to account for the waiting times per lane and includes an ambulance prioritization mechanism. Both systems are evaluated using the SUMO simulator in various traffic scenarios. Performance is assessed through key metrics such as vehicle waiting time and throughput, with results benchmarked against conventional fixed-time control and standalone fuzzy logic approaches.

Despite the extensive body of literature on fuzzy logic for traffic signal control, most existing works focus either on standalone fuzzy systems or on hybrid approaches that combine fuzzy logic with machine learning. In contrast, the present study (see Table I) introduces a novel integration of a Mamdani-type fuzzy inference engine with a gap detection mechanism. This combination enables smarter dynamic adjustment of green-light durations based on real-time traffic data acquired through V2X communication.

III. METHODOLOGY AND SYSTEM DEVELOPMENT

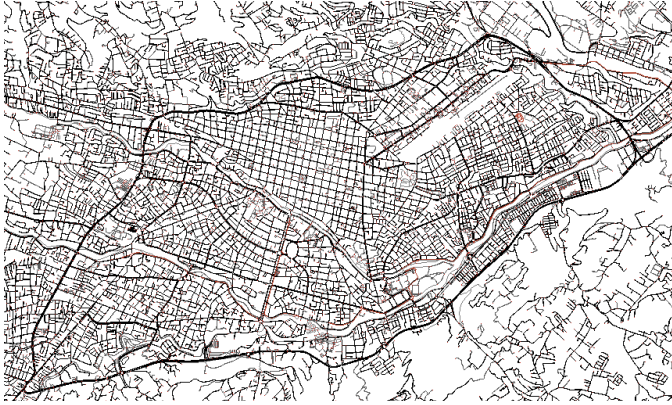
The proposed Gap-Fuzzy control system is assessed in a realistic urban traffic scenario, which is modeled using the SUMO simulation framework. The following subsections describe the study area, the network modeling process, the generation of traffic demand, and the implementation of the traffic signal control strategies used for comparison.

A. Study area and network modeling

The network model was developed using OpenStreetMap (OSM) data and represents a city-scale road network of Cuenca, Ecuador. To enhance realism, the model incorporates detailed traffic signal configurations and turning restrictions. First, lane-specific speed limits consistent with municipal regulations were applied. Second, the city's fixed-time signal

TABLE I: Comparative Summary of Intelligent Traffic Signal Control Approaches

Work	Control Method Used	Input Parameters	Evaluation Metrics	Simulation Tools
[2]	Road network modification	Traffic level	Travel time, speed, waiting time, congestion	SUMO
[3]	Proportional control	Vehicle and pedestrian count using CNN, ANN, and SVM	Waiting time	SUMO, TraCI
[4]	RL	Vehicles' speed and position	CO ₂ emissions, EV energy consumption	SUMO
[5]	Fuzzy-logic	Overflow index, queue length, red phase duration	maximum queue length, vehicle delay	SUMO
[6]	Fuzzy-logic	queue length, waiting time, distance to intersection emergency vehicles	waiting time, fuel consumption, CO ₂	SUMO
[7]	Fuzzy logic	Arrival flow rate	Flow, travel time	MATLAB, Aimsun
[8]	Predictive control	CAV messages for queue and vehicle prediction	Queue length and delay	SUMO
[9]	RL with fuzzy logic	Queue lengths, vehicle count (sensors noisy data)	Throughput (veh/s), number of stops	Real world / Simulated
[10]	Fuzzy logic and Machine Learning (Random Forest + Sugeno approach)	Real-time traffic flow, queue length, and emergency vehicle detection	Total vehicle waiting time, travel time, and traffic flow	SUMO
[11]	Deep Q-Network with fuzzy logic	Average waiting time per lane (lane state)	Total waiting time, per-lane waiting time, and ambulance priority	SUMO
Current Work (Gap-Fuzzy)	Mamdani-type Fuzzy logic and Gap-Out logic	Queue length and arrival rate	Waiting time, CO ₂	SUMO, TraCI

Fig. 2: City-scale SUMO network of Cuenca, Ecuador (urban core; $\approx 114\text{km}^2$ with 361 traffic light systems), derived from OpenStreetMap [12].

plan was integrated for the *static* baseline, instantiating per-phase timings of 42 s green and 6 s amber (yellow) at the signalized junctions. The network encompasses both primary arterial roads and secondary collector / access lanes in the urban core, enabling accurate capture of real-world traffic dynamics. The final network topology is shown in Fig. 2, while the modeling workflow is illustrated in Fig. 6.

To emulate V2X communication, the simulation integrates RSUs in the premises of the traffic control unit and OBUs in the vehicle premises, allowing real-time exchange of vehicle data (see Fig. 1).

B. Traffic-demand generation

We simulate a four-hour horizon under four congestion regimes: low, medium, high, and very high. For each scenario, a target volume-to-capacity ratio $\phi = v/c$ is translated into an hourly vehicle insertion rate (veh/h) as

$$q_{\text{target}} = \phi n_{\text{in}} C_{\ell} \quad [\text{veh/h}], \quad (1)$$

where n_{in} is the effective number of inbound lanes that feed the network boundary and C_{ℓ} is the effective per-lane capacity adjusted for signal density. Over four hours, the implied total demand is $4 q_{\text{target}}$ vehicles per scenario.

These traffic scenarios correspond to typical urban operating regimes and are summarized in Table II. The (i) *low* scenario ($\phi \approx 0.165$) imitates late-night or early-morning conditions in which flows are well below capacity and queues dissipate quickly. The (ii) *medium* scenario ($\phi \approx 0.325$) reflects baseline daytime traffic (e.g., mid-morning or mid-afternoon), where intersections operate stably with moderate queues. Scenario (iii) *high* ($\phi \approx 0.620$) represents peak periods on weekdays with near saturation in critical approaches, longer queues, and increased delay. Finally, the (iv) *very high* scenario ($\phi \approx 0.780$) captures heavy demand approaching 0.8 without network collapse, leading to persistent queues, spillback risk at tightly spaced signals, and pronounced sensitivity to control policies.

C. Signal-control strategies

Three traffic signal control strategies are compared: (i) a fixed-time baseline controller, (ii) the built-in loop-actuated

TABLE II: Scenario targets and implied insertion rates for the Cuenca network.

Scenario	Target $\phi = v/c$	Veh/h	Real-world analogue
low	0.165	1035	Late-night / Early-morning
medium	0.325	2085	Typical daytime (stable)
high	0.620	3860	Weekday peak (near-saturation)
very_high	0.780	4870	Severe peak congestion (near-breakdown)

SUMO controller, and (iii) the proposed Gap-Fuzzy controller. The following subsections offer a detailed description of each control strategy.

1) **Fixed-Time baseline controller:** This configuration replicates the current traffic signal timing implemented in the study area (see Section III-A). Specifically, a fixed 90-second signal cycle is enforced, comprising 42 seconds of green, 6 seconds of amber, and 42 seconds of red, according to municipal traffic regulations. This timing scheme is implemented in the reference SUMO model to ensure consistency with real-world operational conditions.

2) **Loop-Actuated controller:** SUMO's built-in loop-actuated controller adaptively extends green phases as long as vehicle presence is detected by virtual loop detectors. This unmodified algorithm is used as an existing benchmark for adaptive traffic signal control. To ensure a fair and physically plausible comparison with the proposed Gap-Fuzzy strategy, we set the minimum green at 15 s in all actuated approaches (matching the lower bound of the fuzzy controller), fix the amber / yellow interval at 3 s, and limit the maximum green at 50 s. All other parameters follow the defaults of SUMO.

3) **Gap-Fuzzy Controller:** The proposed strategy integrates two complementary control logics: a Mamdani-type fuzzy inference system that estimates the optimal durations of the green phase based on real-time traffic conditions and a gap-out logic that determines the appropriate moment to end the green phase when no vehicles are detected.

First, a detector-actuated safeguard overlay monitors the queue at every simulation step. If (i) the queue remains empty for $T_{\text{gap}} = 2$ s and (ii) the phase has already lasted $T_{\text{min}} = 2$ s, an immediate phase change is triggered, cutting the green short and preventing wasted capacity (see Fig. 3).

Otherwise, the fuzzy decision-making process is triggered. Two input variables are used: queue length and arrival rate, which are first mapped into linguistic fuzzy sets via the fuzzification process. The output is a dynamically adjusted traffic light green duration. Five equal width membership functions (MF) (triangular for interior terms, trapezoidal as shoulders) are defined for each variable. An example of MF for vehicle counts is illustrated in Fig. 4.

To improve predictiveness, both inputs are computed over an approach corridor comprising the lanes mapped to the active phase plus straight-through upstream connections up to two street changes (two links). Internal edges are excluded and

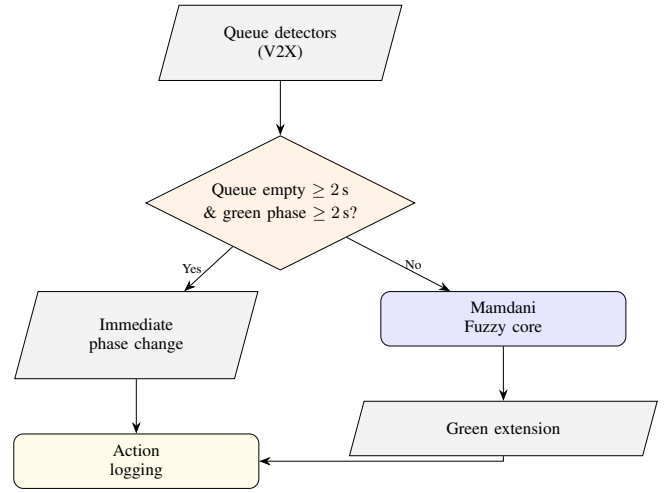


Fig. 3: Block diagram of the gap-out overlay.

contiguous OSM splits along the same street base are treated as one; the queue input sums across corridor lanes, while the arrival-rate input averages nonnegative finite-difference estimates across them.

The internal logic of the proposed Mamdani fuzzy controller is shown in Fig. 5. The process begins with estimating the queue length and arrival rate using beacon messages exchanged through the V2X communication network. If three or fewer vehicles are detected on the beacon table (in the RSU premises), the controller assigns a fixed minimum green time of $l_{\text{min}} = 15$ s to prevent unnecessary delays. Otherwise, the input values are converted to linguistic variables through fuzzification according to the predefined MF.

These fuzzy inputs are then evaluated by an inference engine applying a rule base of 25 rules. A representative excerpt is shown in Table III, and the complete rule base is available in the companion repository [13]. The fuzzy output is defuzzified using the centroid method and is restricted to a practical operating range of 15 to 36 s. The upper bound of 36 s was empirically determined during a simulated annealing (SA) tuning phase: the metaheuristic iteratively adjusted membership interactions and rule consequents to minimize the mean vehicle waiting time, and 36 s consistently achieved the best trade-off (reducing delay without monopolizing an approach). This cap was fixed for all scenarios to ensure comparability and reproducibility.

Note that the fuzzy rules are not made manually but are instead optimized via SA. This metaheuristic systematically adjusts the output values to minimize the average waiting time of the vehicle at the intersection. The optimization runs for a fixed number of iterations (e.g., 40 in the medium-demand scenario), which serves as the sole stopping criterion. Once completed, the resulting rule sets are stored for reproducibility and can be reused in subsequent simulation runs.

Following rule evaluation, the fuzzy output is converted into a crisp green time using the centroid defuzzification method, yielding durations between 15 and 36 seconds. This final

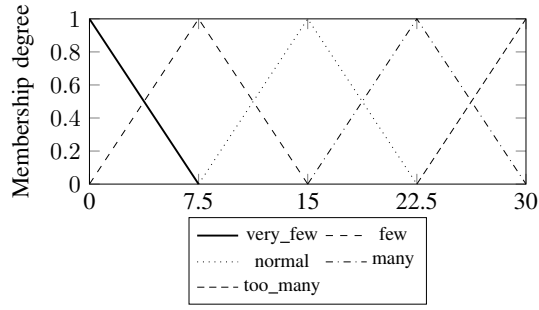


Fig. 4: Equal-width triangular and trapezoidal membership functions for the vehicles input.

TABLE III: Excerpt from the fuzzy rule base (vehicles $\times$ arrival)

Vehicles	Arrival	Green extension
High	Medium	Long
Very High	High	Very Long
Low	Low	Short

decision is applied to the traffic light and logged for post-simulation analysis. The modular structure of the controller enables real-time adaptation of signal timings to dynamic traffic conditions, offering a flexible alternative to a fixed-time control strategy.

This hybrid design preserves the fine-grained adaptation of the fuzzy rules while adding a lightweight gap-out termination mechanism.

Notice that real-time data exchange relies on both the vehicular network (i.e. connected vehicles) and roadside sensing technologies such as cameras and loop detectors. Without loss of generality, vehicle data can originate from the communication network or from sensors deployed along the roadway, making the approach applicable to both current sensor-based road infrastructure and future fully connected transportation systems.

D. Simulation setup

Fig. 6 shows the experimental workflow used to evaluate the traffic control strategies introduced in Section III-C. The process begins with the generation of the network, where the road topology and intersection layout are defined. Then the flow definition, which specifies the OD traffic demands based on Eq. 1. The SUMO simulation stage executes the traffic simulation under one of the three control strategies: Fixed-Time, Loop-Actuated, or the proposed Gap-Fuzzy controller.

After simulation, key performance indicators (KPIs) are extracted (see Fig. 6). KPIs include the mean waiting time for the vehicle (sec / vehicle) and CO₂ emissions (kg / vehicle).

Finally, the statistical analysis stage aggregates and compares each scenario-controller combination evaluated over $N = 10$ Monte Carlo repetitions (i.e., different random seeds), resulting in a total of $4_{\text{scenarios}} \times 3_{\text{strategies}} \times 10_{\text{seeds}} = 120$ simulation runs. All simulations are performed in SUMO with a 1-second simulation step and no warm-up period.

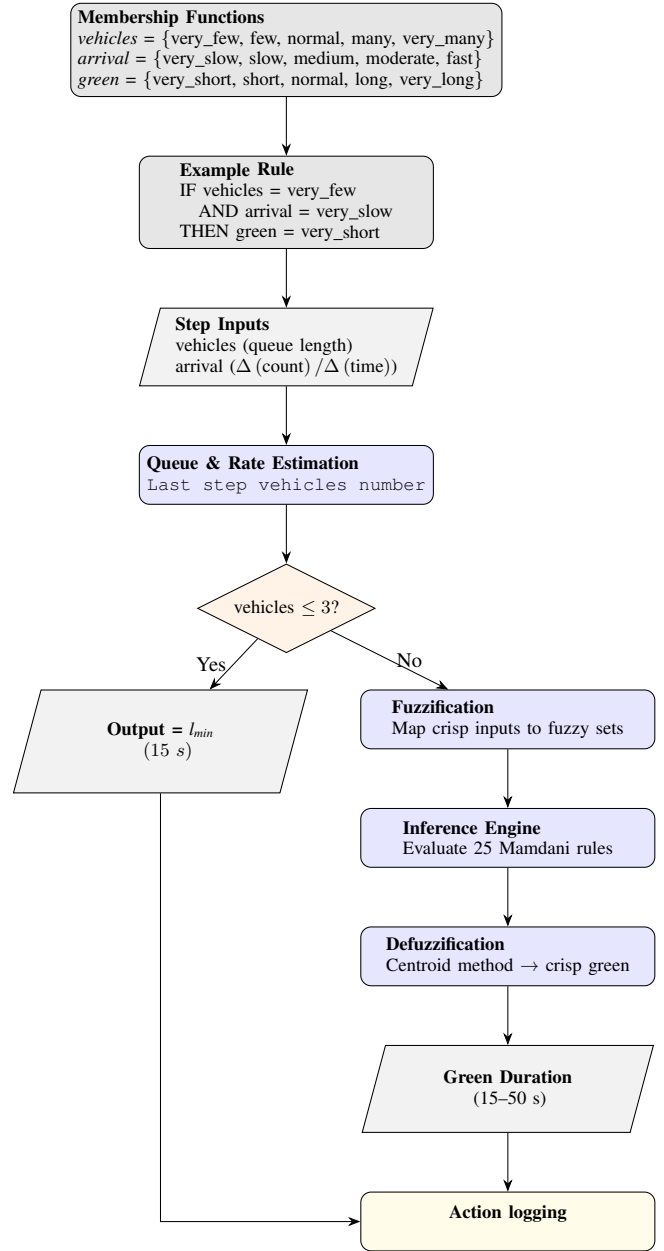


Fig. 5: Block diagram of the Mamdani fuzzy controller.

E. Experimental reproducibility and data access

All experimental artifacts, including network files, pre-generated routes, source code, and aggregated datasets, are openly available at github.com/juanpablo-perez/gap-fuzzy-atasc.

IV. RESULTS

All results are reported with 95% confidence intervals, calculated from the $N = 10$ Monte Carlo simulation repetitions to ensure robust statistical significance and reproducibility. The three control strategies compared are (i) static (*Fixed-Time*), (ii) loop-actuated, and (iii) the proposed Gap-Fuzzy controller.

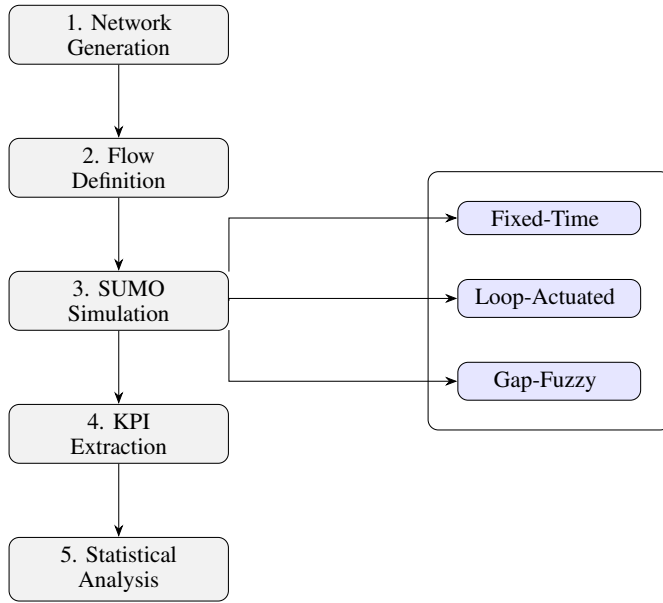


Fig. 6: SUMO experiment pipeline: demand generation, controller injection, and data logging.

A. Per-vehicle mean waiting time

Fig. 7 shows the mean waiting time per vehicle (with 95% CI) for four demand scenarios. Compared to the static baseline, both adaptive controllers offer substantial improvements. Specifically, the loop-actuated controller reduces average waiting times from 65.6% (low demand) to 62.5% (high demand), while the proposed Gap-Fuzzy controller achieves reductions ranging from 76.1% (low) to 61.9% (very high).

A head-to-head comparison between SUMO's built-in actuated controller and Gap-Fuzzy shows:

- **Low demand:** Gap-Fuzzy cuts waiting by an additional **30.6%** relative to actuated, driven by early termination of short-lived queues and corridor-level anticipation.
- **Medium demand:** A further 19.4% reduction, indicating that the fuzzy overlay confers robust gains under moderate and more variable arrivals.
- **High demand:** Gap-Fuzzy still outperforms actuated, but the margin is modest (+2.5%); as approaches near saturation, benefits from gap-out and fuzzy adaptation taper.
- **Very High demand:** Actuated holds a slight edge (1.5% lower mean). Under near-breakdown conditions, queues rarely clear, so gap-out rarely triggers; additionally, the fuzzy cap on green (≤ 36 s) avoids monopolizing an approach but also limits long-clearing greens that presence-based extension can occasionally exploit. The resulting difference remains small.

B. Total CO₂ emissions

All vehicles are fossil-fueled passenger-type vehicles. CO₂ emissions are estimated using the HBEFA 3.1 emission model (PC_G_EU4 category), as implemented in the SUMO traffic simulator [14].

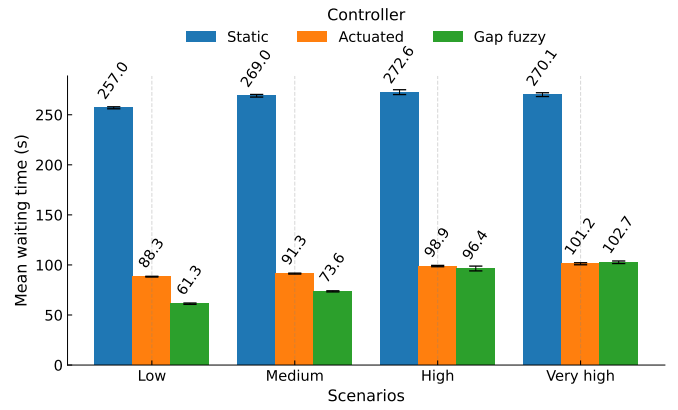


Fig. 7: Mean waiting time per vehicle across four 4-hour demand scenarios. (95% CIs over $N = 10$ runs)

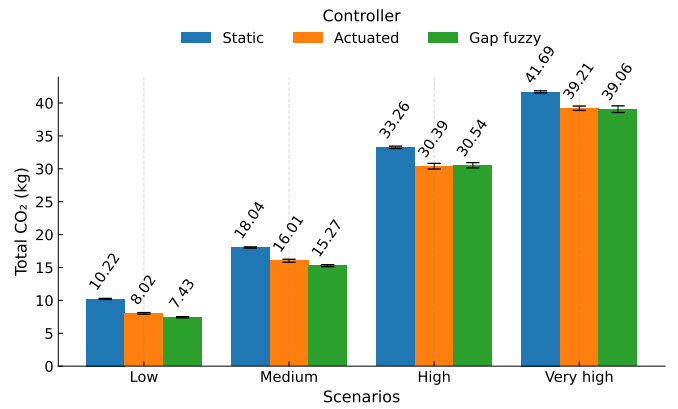


Fig. 8: Total CO₂ emissions [kg] for each control strategy across four 4-hour demand scenarios. (95% CIs over $N = 10$ runs)

Figure 8 shows the total emissions over a 4-hour horizon for all demand scenarios. Both adaptive controllers outperform the fixed-time baseline, with reductions ranging from 6% to 27%. Relative to static, loop-actuated achieves reductions of 21.5%, 11.3%, 8.6%, and 6.0% in low, medium, high and very high traffic demand scenarios, respectively; Gap-Fuzzy achieves 27.3%, 15.4%, 8.2%, and 6.3%. Gap-Fuzzy provides the largest reductions except under high demand, where Loop-Actuated performs slightly better (8.6% vs 8.2%).

Under high demand, this crossover is expected: Loop-Actuated forms steadier platoons with fewer acceleration and braking events, resulting in slightly lower CO₂, while Gap-Fuzzy reduces delay but incurs higher stop-go energy losses.

C. Per-vehicle waiting-time evolution

Figure 9 reports the medium demand profile over a 12h horizon, aggregated in 10-min bins (mean $\pm$ 95% CI). Both adaptive controllers remain consistently below the fixed-time (static) baseline throughout the day. Gap-Fuzzy achieves the lowest waiting times in all intervals, indicating more efficient flow with reduced delays and shorter travel times. The confidence bands are relatively tight across seeds, evidencing stable

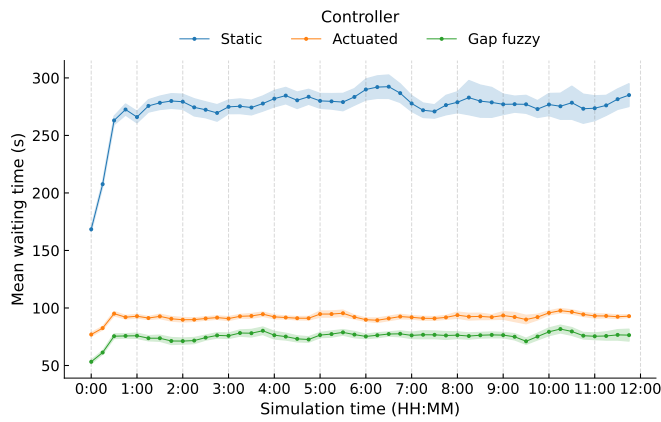


Fig. 9: Mean waiting time per vehicle, 10-min bins over a 12-hour Medium-demand simulation. (95% CIs over $N = 10$ runs)

temporal behavior and reinforcing the aggregate improvements reported in Section IV-A.

V. CONCLUSIONS

This work introduced Gap-Fuzzy, a hybrid traffic signal controller that combines a Mamdani fuzzy regulator (inputs: queue length and arrival rate) with a gap-out mechanism to terminate greens when sustained vehicle gaps occur.

City-scale experiments in Cuenca (Ecuador) in four demand regimes show that Gap-Fuzzy consistently outperforms the fixed-time (Static) baseline and is competitive with SUMO's Loop Actuated controller. In mean waiting time, Gap-Fuzzy delivers large absolute gains over Static (about 60–75%) and improves on Loop-Actuated at Low/Medium demand, with a small edge at High; under near-breakdown (Very High), the actuated controller leads. For emissions, both adaptive strategies reduce CO₂ relative to Static by $\sim 6\%$ –27%. Gap-Fuzzy achieves the largest cuts at Low/Medium (and remains competitive at Very High), whereas at High demand Loop-Actuated is marginally better—consistent with smoother trajectories (fewer stops) near saturation.

These reductions highlight the effectiveness of adaptive fuzzy logic combined with gap-out logic in minimizing intersection delays, particularly under moderate traffic loads. These results suggest that the Gap-Fuzzy approach can accommodate higher traffic volumes without causing network deadlocks, thus demonstrating better adaptability to dynamic traffic conditions.

Future work will explore (i) automated, scenario-specific tuning of membership functions and consequents (e.g., via simulated annealing or Bayesian optimization), and (ii) adaptive learning of gap thresholds through reinforcement learning under safety constraints.

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